Karatina University

STA 205:Probability & Statistics ASSIGNMENT (25 marks) Time:1hr

Instructions: Answer all Questions

1. The number of claims handled daily by an insurance company is assumed to be Poisson distributed with parameter $\lambda = 5$. Calculate the proportion of days that had less than 3 claims. [3 marks]
2. The masses of 300 students are normally distributed with mean $68kg$ and standard deviation $3kg$. Determine the number of students whose mass is greater than $72kg$. [3 marks]
3. The distribution of heights of students in a class follows a normal distribution with a mean of 160cm and a standard deviation of 6.2 cm. Find the probability that a student selected at random has a height of more than 155 cm. [3 marks]
4. It is known that 10 percent of the chips produced by a computer hardware manufacturer are defective. An institution orders 20 chips. Determine the probability that at most 3 chips were defective. [3 marks]
5. A cybersecurity company is testing a new intrusion detection system (IDS) that identifies malicious login attempts. Based on historical data, the probability that the IDS correctly detects a malicious login attempt is 0.75. The company tests the system on 10 independent malicious login attempts. What is the probability that the IDS correctly detects at least 9out of the 10 malicious attempts? [3 marks]
6. Two computer science students, A and B are working on a new project to develop an AI-based code completion tool. They track the time (in minutes) taken to complete 10 coding tasks individually. The collected data is as follows:

	A	B
Number of tasks	15	15
Average time(in minutes)	28	30
Standard deviation(in minutes)	1.5	1.0

Test whether there is any significant difference in the average time between the two students(state the null and alternative hypothesis then test). [4 marks]

7. A software company wants to determine if there is a relationship between the number of lines of code and the time (in hours) taken to develop a software module. The company collected data from 10 recently completed projects, as shown below:

Project	1	2	3	4	5	6	7	8	9	10
Lines of Code (X)	20	45	35	55	70	65	40	55	30	50
Development Time (Hours) (Y)	1	2.5	1.5	3	5	4.5	2	3.5	2	4

a) Determine the regression equation. [4 marks]

b) Calculate the Pearson's correlation coefficient (r) and interpret its value. [2 marks]

Practice Questions

a) Revise on Discrete and continuous random variables, their characteristics and how to find expectation, variance and probability.

b) Distinguish between the following terms as used in statistics:

- A parameter and a statistic.
- Type I error and Type II error.
- Regression and Correlation analysis

c) Binomial and Poisson are both probability distributions. Describe the similarities between the two distributions.

d) Explain the general procedure followed in testing hypothesis

e) <u>RESEARCH QUESTION ON USING SPSS TO ANALYSE DATA AND THEN ANSWER THE FOLLOWING QUESTION.</u>

Two different types of drugs A and B were tried on certain patients for increasing weight, 5 persons were given drug A and 7 persons were given drug B. The increase in weight (in pounds) is given below:

Drug A	8	12	13	9	3		
Drug B	10	8	12	15	6	8	11

The analysis of the results was done using **SPSS** and the following output was obtained.

Independent Samples Test

	Levene's Test for Equality of Variances		t-test for Equality of Means						
	F	Sig.	T	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
								Lower	Upper
Equal variances assumed	.192	.671	-.501	10	.627	-1.000	1.994	-5.444	3.444
Equal variances not assumed			-.478	7.182	.647	-1.000	2.094	-5.927	3.927

By stating the appropriate null and alternative hypotheses, the decision criteria and test statistic, use the provided **SPSS** output above to determine whether the two drugs differ significantly with regard to their effect in increasing weight.